

4. Exercise: Deployment and optimization

name(s):	points:	(max. 24)
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In this exercise, you will infer your created gesture-recognition model from Exercise 3 on the Raspberry Pi. You will deploy the baseline model, the pruned models, and the quantized models to the device and run inference with a local CSV time-series sample. The goal is to show that all required models can be converted, deployed, and used for inference on the Raspberry Pi.

- The code must be **your own work**.
- You may use online resources to understand TensorFlow, TFLite, pruning and quantization.
- You must not use generative AI tools to write the code for you.
- You must be able to explain your code and design decisions in the lesson.
- **Failure to understand / explain** your own code and design decisions can lead to **negative grading**.

1.Part *Prepare one repeatable test sample.*

3 points

- (a) Select a single CSV file from your gesture dataset and copy it to the artifact directory in your development container. This file should contain one representative time-series example that can be used for repeated inference on the Raspberry Pi.

The existing deployment script already contains a segment that copies the MNIST test image to the Raspberry Pi:

```
if [[ -f "${ARTIFACT_DIR}/test_digit.bmp" ]]; then
    ARTIFACT_FILES+=("${ARTIFACT_DIR}/test_digit.bmp")
fi
```

- (b) Duplicate this segment in your *deploy_to_pi.sh* and modify it so that your gesture CSV file is also deployed.

2.Part *Add your classifier.*

3 points

- (a) Instead of a *TfliteDigitClassifier*, your application should use a *TfliteGestureClassifier*. The classifier should support a four-class gesture model. The prediction result should contain at least:

- the predicted class index,

- the confidence,
- optionally the full score vector.

You should use the internal helper functions that already handle the supported tensor formats, especially FP32 and INT8.

You can orient your implementation on the existing *TfliteDigitClassifier*.

3. Part *Define the CSV input file in main.cpp.*

3 points

- (a) Define your CSV file inside *main.cpp*, for example:

```
constexpr const char* kCsvPath = "test_gesture.csv";
```

Your program can also support a command-line option such as:

```
--csv test_gesture.csv
```

- (b) The program should load this CSV file and convert it into the flat time-series input expected by your model.

4. Part *Modify the inference flow in main.cpp.*

4 points

- (a) Modify the provided inference function so that it uses *ProgramOptions* and *TfliteGestureClassifier* instead of the old digit classifier. The function should:

- load the selected CSV file,
- create the model input vector,
- call *classifier->Predict(...)* or *classifier->predict(...)*,
- print the predicted gesture class,
- print the confidence or class scores.

You can use the current implementation of the digit classifier inference flow as a template, but remove all parts that are only relevant for images, camera capture, MNIST preprocessing or Sense HAT digit display.

5. Part *Create the required TFLite models.*

5 points

- (a) Your baseline model should be your four-class gesture-recognition model. The pruned and quantized models should be derived from this baseline model. Use your training and conversion pipeline from the class to create

the following models:

```
baseline.tflite
synapse_pruned.tflite
neuron_pruned.tflite
channel_pruned.tflite
baseline_int8.tflite
```

6. Part *Deploy all models to the Raspberry Pi.*

2 points

- (a) Update your deployment script so that all required `.tflite` models are copied to the Raspberry Pi.

The remote directory should contain all models from **5. Part**.

7. Part *Run inference on the Raspberry Pi.*

4 points

- (a) Run your application once for each required model. You can use the provided tasks from within your development container or optionally run commands on the Pi (with a program named `gesture_demo`) such as:

```
./gesture_demo --model baseline.tflite --csv test_gesture.csv
./gesture_demo --model synapse_pruned.tflite --csv test_gesture.csv
./gesture_demo --model neuron_pruned.tflite --csv test_gesture.csv
./gesture_demo --model channel_pruned.tflite --csv test_gesture.csv
./gesture_demo --model baseline_int8.tflite --csv test_gesture.csv
```

Command line arguments are only required, if you implemented them in **3. Part**